Chapter 19
The Least Squares Problem

For some time it has been believed that orthogonalizing methods
did not suffer this squaring of the condition number . . .
It caused something of a shock, therefore,
when in 1966 *Golub and Wilkinson . . . asserted that*
already the multiplications QA and Qb may produce errors in the solution
containing a factor $x^2(A)$.

— A. VAN DER SLUIS,
Stability of the Solutions of Linear Least Squares Problems (1975)

Most packaged regression problems do compute a cross-products matrix
and solve the normal equations using a matrix inversion subroutine.
All the programs . . . that disagreed
(and some of those that agreed) with the unperturbed solution
tried to solve the normal equations.

— ALBERT E. BEATON, DONALD B. RUBIN, and JOHN L. BARONE,
The Acceptability of Regression Solutions:
Another Look at Computational Accuracy (1976)

On January 1, 1801 *Giuseppe Piazzi discovered the asteroid Ceres.*
Ceres was only visible for forty days
before it was lost to view behind the sun . . .
Gauss, using three observations, extensive analysis,
and the method of least squares, was able to
determine the orbit with such accuracy that Ceres was
easily found when it reappeared in late 1801.

— DAVID K. KAHANER, CLEVE B. MOLER, and STEPHEN G. NASH,
Numerical Methods and Software (1989)

In this chapter we consider the least squares (LS) problem $\min_x \|b - Ax\|_2$, where $A \in \mathbb{R}^{m\times n}$ ($m > n$) has full rank. We begin by examining the sensitivity of the LS problem to perturbations. Then we examine the stability of methods for solving the LS problem, covering QR factorization methods, the normal equations and seminormal equations methods, and iterative refinement. Finally, we show how to compute the backward error of an approximate LS solution.

We do not develop the basic theory of the LS problem, which can be found in standard textbooks (see, for example, Golub and Van Loan [470, 1989, §5.3]). However, we recall the fundamental result that any solution of the LS problem satisfies the normal equations $A^TAx = A^Tb$ (see Problem 19.1). Therefore if A has full rank there is a unique LS solution. More generally, whatever the rank of A the vector $x_{LS} = A^+b$ is an LS solution, and it is the solution of minimal 2-norm. Here, A^+ is the pseudo-inverse of A (given by $A^+ = (A^TA)^{-1}A^T$ when A has full rank); see Problem 19.3. (For more on the pseudo-inverse see Stewart and Sun [954, 1990, §3.1].)

19.1. Perturbation Theory

Perturbation theory for the LS problem is, not surprisingly, more complicated than for linear systems, and there are several forms in which bounds can be stated. We begin with a normwise perturbation theorem that is a restatement of a result of Wedin [1069, 1973, Thm. 5.1]. For an arbitrary rectangular matrix A we define the condition number $\kappa_2(A) = \|A\|_2\|A^+\|_2$. If A has $r = \operatorname{rank}(A)$ nonzero singular values, $\sigma_1 \ge \cdots \ge \sigma_r$, then $\kappa_2(A) = \sigma_1/\sigma_r$.

Theorem 19.1 (Wedin). *Let $A \in \mathbb{R}^{m\times n}$ ($m \ge n$) and $A + \Delta A$ both be of full rank, and let*

$$\|b - Ax\|_2 = \min, \qquad r = b - Ax,$$
$$\|(b+\Delta b) - (A+\Delta A)y\|_2 = \min, \qquad s = b + \Delta b - (A+\Delta A)y,$$
$$\|\Delta A\|_2 \le \epsilon\|A\|_2, \quad \|\Delta b\|_2 \le \epsilon\|b\|_2.$$

Then, provided that $\kappa_2(A)\epsilon < 1$,

$$\frac{\|x-y\|_2}{\|x\|_2} \le \frac{\kappa_2(A)\epsilon}{1-\kappa_2(A)\epsilon}\left(2 + (\kappa_2(A)+1)\frac{\|r\|_2}{\|A\|_2\|x\|_2}\right) \tag{19.1}$$

$$\frac{\|r-s\|_2}{\|b\|_2} \le (1 + 2\kappa_2(A))\epsilon. \tag{19.2}$$

These bounds are approximately attainable. □

Proof. We defer a proof until § 19.8, since the techniques used in the proof are not needed elsewhere in this chapter. □

The bound (19. 1) is usually interpreted as saying that the sensitivity of the LS problem is measured by $\kappa_2(A)$ when the residual is small or zero and by $\kappa_2(A)^2$ otherwise. This means that the sensitivity of the LS problem depends strongly on b as well as A, unlike for a square linear system.

Here is a simple example where the $\kappa_2(A)^2$ effect is seen:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & \epsilon \\ 0 & 0 \end{bmatrix}, \quad \Delta A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & \epsilon \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \Delta b = 0.$$

It is a simple exercise to verify that

$$x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad r = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad y = \begin{bmatrix} 1 \\ \frac{1}{2\epsilon} \end{bmatrix}, \quad s = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Since $\kappa_2(A) = 1/\varepsilon$,

$$\frac{\|x-y\|_2}{\|x\|_2} = \frac{1}{2\epsilon} \lesssim \kappa_2(A)^2 \frac{\|\Delta A\|_2}{\|A\|_2} = \frac{1}{\epsilon},$$
$$\frac{\|r-s\|_2}{\|b\|_2} = \frac{1}{2} \lesssim \kappa_2(A) \frac{\|\Delta A\|_2}{\|A\|_2} = 1.$$

Surprisingly, it is easier to derive componentwise perturbation bounds than normwise ones for the LS problem. The key idea is to express the LS solution and its residual as the solution of the augmented system

$$\begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}, \tag{19.3}$$

which is simply another way of writing the normal equations, $A^TAx = A^Tb$. This is a square nonsingular system, so standard techniques can be applied. The perturbed system of interest is

$$\begin{bmatrix} I & A+\Delta A \\ (A+\Delta A)^T & 0 \end{bmatrix} \begin{bmatrix} s \\ y \end{bmatrix} = \begin{bmatrix} b+\Delta b \\ 0 \end{bmatrix}, \tag{19.4}$$

where we assume that

$$|\Delta A| \le \epsilon E, \quad |\Delta b| \le \epsilon f. \tag{19.5}$$

From (19.3) and (19.4) we obtain

$$\begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} s-r \\ y-x \end{bmatrix} = \begin{bmatrix} \Delta b - \Delta A y \\ -\Delta A^T s \end{bmatrix}.$$

Premultiplying by the inverse of the matrix on the left gives

$$\begin{bmatrix} s-r \\ y-x \end{bmatrix} = \begin{bmatrix} I-AA^+ & (A^+)^T \\ A^+ & -(A^TA)^{-1} \end{bmatrix} \begin{bmatrix} \Delta b - \Delta A y \\ -\Delta A^T s \end{bmatrix}. \tag{19.6}$$

Looking at the individual block components we obtain

$$|s-r| \le \epsilon\big(|I-A^+A|(f+E|y|) + |A^+|^T E^T |s|\big), \tag{19.7}$$

$$|y-x| \le \epsilon\big(|A^+|(f+E|y|) + |(A^TA)^{-1}|E^T|s|\big). \tag{19.8}$$

(Note that $\|I - AA^+\|_2 = \min\{1, m-n\}$, as is easily proved using the singular value decomposition (SVD).) On taking norms we obtain the desired perturbation result.

Theorem 19.2. *Let $A \in \mathbb{R}^{m\times n}$ $(m \ge n)$ and $A + \Delta A$ be of full rank. For the perturbed LS problem described by (19.4) and (19.5) we have*

$$\frac{\|x-y\|}{\|x\|} \le \epsilon \frac{\|\,|A^+|(f+E|y|)\,\| + \|\,|(A^TA)^{-1}|E^T|s|\,\|}{\|x\|}, \tag{19.9}$$

$$\frac{\|r-s\|}{\|r\|} \le \epsilon \frac{\|\,|I-AA^+|(f+E|y|)\,\| + \|\,|A^+|^T E^T|s|\,\|}{\|r\|}, \tag{19.10}$$

for any monotonic norm. These bounds are approximately attainable. □

For a square system, we have $s = 0$, and we essentially recover Theorem 7.4. Note, however, that the bounds contain the perturbed vectors y and s. For theoretical analysis it may be preferable to use alternative bounds in which x and r replace y and s and there is an extra factor

$$\left(1 - \epsilon\left\|\begin{bmatrix} |I-AA^+| & |A^+|^T \\ |A^+| & |(A^TA)^{-1}| \end{bmatrix}\begin{bmatrix} 0 & E \\ E^T & 0 \end{bmatrix}\right\|\right)^{-1},$$

where the term in parentheses is assumed to be positive. For practical computation (19.9) is unsatisfactory because we do not know $s = b + \Delta b - (A + \Delta A)y$. However, as Stewart and Sun observe [954, 1990, p. 159], $\widehat{r} = b - Ay$ is computable and

$$|s| \le |\widehat{r}| + \epsilon(f + E|y|),$$

and using this bound in (19.9) makes only a second-order change.

The componentwise bounds enjoy better scaling properties than the normwise ones. If $E = |A|$ and $f = |b|$ then the bounds (19.7) and (19.8), and to a lesser extent (19.9) and (19.10), are invariant under column scalings $b - Ax \to b - AD \cdot D^{-1}x$ (D diagonal). Row scaling does affect the componentwise bounds, since it changes the LS solution, but the componentwise bounds are less sensitive to the row scaling than the normwise bounds, in a way that is difficult to make precise.

19.2. Solution by QR Factorization

Let $A \in \mathbb{R}^{m\times n}$, with $m \geq n$ and rank$(A) = n$. If A has the QR factorization

$$Q^T A = \begin{bmatrix} R \\ 0 \end{bmatrix}$$

then

$$\|Ax - b\|_2^2 = \|Q^T Ax - Q^T b\|_2^2 =: \left\| \begin{bmatrix} Rx - c \\ -d \end{bmatrix} \right\|_2^2$$

It follows that the unique LS solution is $x = R^{-1}c$, and the residual $\|b - Ax\|_2 = \|d\|_2$. Thus the LS problem can be solved with relatively little extra work beyond the computation of a QR factorization. Note that Q is not required explicitly; we just need the ability to apply Q^t to a vector.

It is well known that the Givens and Householder QR factorization algorithms provide a normwise backward stable way to solve the LS problem. The next result expresses this fact for the Householder method and also provides componentwise backward error bounds (essentially the same result holds for the Givens method).

As in Chapter 18, we will use the generic constant γ_{cm}, in which c denotes a small integer. We will assume implicitly that a condition holds of the form $mn\gamma_{cm} < 1/2$.

Theorem 19.3. *Let $A \in \mathbb{R}^{m\times n}$ $(m \geq n)$ have full rank and suppose the LS problem $\min_x \|b - Ax\|_2$ is solved using the Householder QR factorization method. The computed solution $\widehat{x}$ is the exact LS solution to*

$$\min_x \|(b + \Delta b) - (A + \Delta A)\widehat{x}\|_2,$$

where the perturbations satisfy the normwise bounds

$$\|\Delta A\|_F \leq n\gamma_{cm}\|A\|_F, \quad \|\Delta b\|_2 \leq n\gamma_{cm}\|b\|_2,$$

and the componentwise bounds

$$|\Delta A| \leq mn\gamma_{cm} G|A|, \qquad |\Delta b| \leq mn\gamma_{cm} G|b|,$$

where $\|G\|_F = 1$.

Proof. The proof is a straightforward generalization of the proof of Theorem 18.5 and is left as an exercise (Problem 19.2). $\square$

As for Theorem 18.5 (see (18.7)), Theorem 19.3 remains true if we set $\Delta b \equiv 0$, but in general there is no advantage to restricting the perturbations to A.

Theorem 19.3 is a strong result, but it does not bound the residual of the computed solution, which, after all, is what we are trying to minimize. How close, then, is $\|b-A\widehat{x}\|_2$ to $\min_x \|b - Ax\|_2$? We can answer this question using the perturbation theory of §19.1. With $\widehat{r} := b + \Delta b - (A + \Delta A)\widehat{x}$, $x := x_{LS}$ and $r := b - Ax$, (19.6) yields

$$\widehat{r} - r = (I - AA^+)(\Delta b - \Delta A\widehat{x}) - (A^+)^T \Delta A^T \widehat{r},$$

so that

$$(b - A\widehat{x}) - r = -AA^+(\Delta b - \Delta A\widehat{x}) - (A^+)^T \Delta A^T \widehat{r}.$$

Substituting the bounds for ΔA and Δb from Theorem 19.3, and noting that $\|AA^+\|_2 = 1$, we obtain

$$\begin{aligned}\|(b - A\widehat{x}) - r\|_2 &\le mn\gamma_{cm}\big(\|G(|b| + |A||x|)\|_2 \\ &\qquad + \|\,|A^+|^T|A^T|G^T|r|\,\|_2\big) + O(u^2), \\ &\le mn\gamma_{cm}\big(\|\,|b| + |A||x|\,\|_2 + \mathrm{cond}_2(A^T)\|r\|_2\big) + O(u^2),\end{aligned}$$

where $\mathrm{cond}_2(A) := \|\,|A^+||A|\,\|_2$. Hence

$$\|b - A\widehat{x}\|_2 \le mn\gamma_{cm}\|\,|b| + |A||x|\,\|_2 + \big(1 + mn\gamma_{cm}\,\mathrm{cond}_2(A^T)\big)\|r\|_2 + O(u^2).$$

This bound contains two parts. The term $mn\gamma_{cm}\|\,|b| + |A||x|\,\|_2$ is a multiple of the bound for the error in evaluating $fl(b - Ax)$, and so is to be expected. The factor $1 + mn\gamma_{cm}\,\mathrm{cond}_2(A^T)$ will be less than 1.1 (say) provided that $\mathrm{cond}_2(A^T)$ is not too large. Note that $\mathrm{cond}_2(A^T) \le \kappa_2(A)$ and $\mathrm{cond}_2(A^T)$ is invariant under column scaling of A ($A \leftarrow A\,\mathrm{diag}(d_i)$, $d_i \ne 0$). The conclusion is that, unless A is very ill conditioned, the residual $b - A\widehat{x}$ will not exceed the larger of the true residual $r = b - Ax$ and a constant multiple of the error in evaluating $fl(r)$—a very satisfactory result.

19.3. Solution by the Modified Gram–Schmidt Method

The modified Gram-Schmidt (MGS) method can be used to solve the LS problem. However, we must not compute x from $x = R^{-1}(Q^T b)$, because the lack of orthonormality of the computed $\widehat{Q}$ would adversely affect the stability. Instead we apply MGS to the augmented matrix $[A \;\; b]$:

$$[A \;\; b] = [Q_1 \;\; q_{n+1}] \begin{bmatrix} R & z \\ 0 & \rho \end{bmatrix}.$$

We have

$$\begin{aligned} Ax - b = [A \ \ b]\begin{bmatrix} x \\ -1 \end{bmatrix} &= [Q_1 \ \ q_{n+1}]\begin{bmatrix} Rx - z \\ -\rho \end{bmatrix} \\ &= Q_1(Rx - z) - \rho q_{n+1}. \end{aligned}$$

Since q_{n+1} is orthogonal to the columns of Q_1, $\|b - Ax\|_2^2 = \|Rx - z\|_2^2 + \rho^2$, so the LS solution is $x = R^{-1}z$. Of course, $z = Q_1^T b$, but z is now computed as part of the MGS procedure instead of as a product between Q^T and b

Björck [107, 1967] shows that this algorithm is forward stable, in the sense that the forward error $\|x - \widehat{x}\|_2/\|x\|_2$ is as small as that for a normwise backward stable method. It has recently been shown by Björck and Paige [119, 1992] that the algorithm is, in fact, normwise backward stable (see also Björck [115, 1994]), that is, a normwise result of the form in Theorem 19.3 holds. Moreover, a componentwise result of the form in Theorem 19.3 holds too—see Problem 19.5. Hence the possible lack of orthonormality of $\widehat{Q}$ does not impair the stability of the MGS method as a means for solving the LS problem.

19.4. The Normal Equations

The oldest method of solving the LS problem is to form and solve the normal equations, $A^TAx = A^Tb$. Assuming that A has full rank, we can use the following procedure:

Form $C = A^TA$ and $c = A^Tb$.
Compute the Cholesky factorization $C = R^TR$.
Solve $R^Ty = c$, $Rx = y$.

Cost: : $n^2(m + n/3)$ flops.

If $m >> n$, the normal equations method requires about half as many flops as the Householder QR factorization approach (or the MGS method). However, it has less satisfactory numerical stability properties. There are two problems. The first is that information may be lost when $\widehat{C} := fl(A^TA)$ is formed-essentially because forming the cross product is a squaring operation that increases the dynamic range of the data. A simple example is the matrix

$$A = \begin{bmatrix} 1 & 1 \\ \epsilon & 0 \end{bmatrix}, \qquad 0 < \epsilon < \sqrt{u},$$

for which

$$A^TA = \begin{bmatrix} 1+\epsilon^2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad fl(A^TA) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Even though A is distance approximately ε from a rank-deficient matrix, and hence unambiguously full rank if $\varepsilon \approx \sqrt{u/2}$, the computed cross product is singular. In general, whenever $\kappa_2(A) \geq u^{-1/2}$ we can expect $\widehat{C}$ to be singular or indefinite, in which case Cholesky factorization is likely to break down (Theorem 10.7).

The second weakness of the normal equations method is more subtle and is explained by a rounding error analysis. In place of $C = A^TA$ and $c = A^Tb$ we compute

$$\begin{aligned} \widehat{C} &= A^TA + \Delta C_1, \qquad |\Delta C_1| \leq \gamma_m |A^T||A|, \\ \widehat{c} &= A^Tb + \Delta c, \qquad |\Delta c| \leq \gamma_m |A^T||b|. \end{aligned}$$

By Theorems 10.3 and 10.4, the computed Cholesky factor $\widehat{R}$ and solution $\widehat{x}$ satisfy

$$\widehat{R}^T\widehat{R} = \widehat{C} + \Delta C_2, \qquad |\Delta C_2| \leq \gamma_{n+1}|\widehat{R}^T||\widehat{R}|, \tag{19.11}$$
$$(\widehat{C} + \Delta C_3)\widehat{x} = \widehat{c}, \qquad |\Delta C_3| \leq 2\gamma_{n+1}|\widehat{R}^T||\widehat{R}|.$$

Overall,

$$(A^TA + \Delta A)\widehat{x} = A^Tb + \Delta c, \tag{19.12}$$
$$|\Delta A| \leq \gamma_m|A^T||A| + 2\gamma_{n+1}|\widehat{R}^T||\widehat{R}|, \quad |\Delta c| \leq \gamma_m|A^T||b|.$$

By bounding $\| |\widehat{R}^T||\widehat{R}| \|_2$ with the aid of (19.11), we find that

$$\|\Delta A\|_2 \leq (mn + 2n^2 + 2n)u\|A\|_2^2 + O(u^2), \tag{19.13a}$$
$$\|\Delta c\|_2 \leq mn^{1/2}u\|A\|_2\|b\|_2 + O(u^2). \tag{19.13b}$$

These bounds show that we have solved the normal equations in a backward stable way, as long as $\|A\|_2\|b\|_2 \approx \|A^Tb\|_2$. But if we try to translate this result into a backward error result for the LS problem itself, we find that the best backward error bound contains a factor $\kappa_2(A)$ [569, 1987]. The best forward error bound we can expect, in view of (19.13), is of the form

$$\frac{\|x - \widehat{x}\|_2}{\|x\|_2} \lesssim c_{m,n}\kappa_2(A)^2u \tag{19.14}$$

(since $\kappa_2(A^TA) = \kappa_2(A)^2$). Now we know from Theorem 19.1 that the sensitivity of the LS problem is measured by $\kappa_2(A)^2$ if the residual is large, but by $\kappa_2(A)$ if the residual is small. It follows that the normal equations method has a forward error bound that can be much larger than that possessed by a backward stable method.

A mitigating factor for the normal equations method is that, in view of Theorem 10.6, we can replace (19.14) by the (not entirely rigorous) bound

$$\frac{\|D(x - \widehat{x})\|_2}{\|Dx\|_2} \lesssim c_{m,n}\kappa_2(B)^2u,$$

where $A = BD$, with $D = \mathrm{diag}(\|A(:, i)\|_2)$, so that B has columns of unit 2-norm. Van der Sluis's result (Theorem 7.5) shows that

$$\kappa_2(B) \le \sqrt{n} \min_{F \text{ diagonal}} \kappa_2(AF).$$

Hence the normal equations method is, to some extent, insensitive to poor column scaling of A.

Although numerical analysts almost invariably solve the full rank LS problem by QR factorization, statisticians frequently use the normal equations (though perhaps less frequently than they used to, thanks to the influence of numerical analysts). The normal equations do have a useful role to play. In many statistical problems the regression matrix is contaminated by errors of measurement that are very large relative to the roundoff level; the effects of rounding errors are then likely to be insignificant compared with the effects of the measurement errors, especially if IEEE double precision (as opposed to single precision) arithmetic is used.

The normal equations (NE) versus (Householder) QR factorization debate can be summed up as follows.

- The two methods have a similar computational cost if $m \approx n$, but the NE method is up to twice as fast for $m >> n$. (This statement assumes that A and b are dense; for details of the storage requirements and computational cost of each method for sparse matrices, see, for example, Björck [116, 1996] and Heath [510, 1984].)
- The QR method is always backward stable. The NE method is guaranteed to be backward stable only if A is well conditioned.
- The forward error from the NE method can be expected to exceed that for the QR method when A is ill conditioned and the residual of the LS problem is small.
- The QR method lends itself to iterative refinement, as described in the next section. Iterative refinement can be applied to the NE method, but the rate of convergence inevitably depends on $\kappa_2(A)^2$ instead of $\kappa_2(A)$.

19.5. Iterative Refinement

As for square linear systems, iterative refinement can be used to improve the accuracy and stability of an approximate LS solution. However, for the LS problem there is more than one way to construct an iterative refinement scheme.

By direct analogy with the square system case, we might consider the scheme

1. Compute $r = b - A\widehat{x}$.
2. Solve the LS problem $\min_d \|Ad - r\|_2$.
3. Update $y = \widehat{x} + d$.

   (Repeat from step 1 if necessary, with $\widehat{x}$ replaced by y).

This scheme is certainly efficient-a computed QR factorization (for example) of A can be reused at each step 2. Golub and Wilkinson [466, 1966] investigated this scheme and found that it works well only for nearly consistent systems.

An alternative approach suggested by Björck [106, 1967] is to apply iterative refinement to the augmented system (19.3), so that both x and r are refined simultaneously. Since this is a square, nonsingular system, existing results on the convergence and stability of iterative refinement can be applied, and we would expect the scheme to work well. To make precise statements we need to examine the augmented system method in detail.

For the refinement steps we need to consider an augmented system with an arbitrary right-hand side:

$$r + Ax = f, \tag{19.15a}$$

$$A^T r = g. \tag{19.15b}$$

If A has the QR factorization

$$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix},$$

where $R \in \mathbb{R}^{n \times n}$, then (19. 15) transforms to

$$Q^T r + \begin{bmatrix} R \\ 0 \end{bmatrix} x = Q^T f,$$

$$[R^T \quad 0] Q^T r = g.$$

This system can be solved as follows:

$$h = R^{-T} g,$$

$$d = Q^T f = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix},$$

$$r = Q \begin{bmatrix} h \\ d_2 \end{bmatrix},$$

$$x = R^{-1}(d_1 - h).$$

The next result describes the effect of rounding errors on the solution process.

Theorem 19.4. *Let $A \in \mathbb{R}^{m\times n}$ be of full rank $n \le m$ and suppose the augmented system* (19.15) *is solved using a Householder QR factorization of A as described above. The computed $\widehat{x}$ and $\widehat{r}$ satisfy*

$$\begin{bmatrix} I & A+\Delta A_1 \\ (A+\Delta A_2)^T & 0 \end{bmatrix} \begin{bmatrix} \widehat{r} \\ \widehat{x} \end{bmatrix} = \begin{bmatrix} f+\Delta f \\ g+\Delta g \end{bmatrix},$$

where

$$\begin{aligned} |\Delta A_i| &\le mn\gamma_{cm} G|A|, \qquad i = 1\colon 2, \\ |\Delta f| &\le \sqrt{m}n\gamma_{cm}(H_1|f| + H_2|\widehat{r}|), \\ |\Delta g| &\le \sqrt{m}n\gamma_{cm}|A^T|H_3|\widehat{r}|, \end{aligned}$$

with $\|G\|_F = 1$, $\|H_i\|_F = 1$, $i = 1{:}3$.

Proof. The proof does not involve any new ideas and is rather tedious, so we omit it. □

Consider first fixed precision iterative refinement. Theorem 19.4 implies that the computed solution $(\widehat{r}^T, \widehat{x}^T)^T$ to (19.15) satisfies

$$\left| \begin{bmatrix} f \\ g \end{bmatrix} - \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} \widehat{r} \\ \widehat{x} \end{bmatrix} \right| \le mn\gamma_{cm} \left(\begin{bmatrix} H_2 & G|A| \\ |A^T|(G^T + H_3) & 0 \end{bmatrix} \begin{bmatrix} |\widehat{r}| \\ |\widehat{x}| \end{bmatrix} + \begin{bmatrix} H_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} |f| \\ |g| \end{bmatrix} \right).$$

Unfortunately, this bound is not of a form that allows us to invoke Theorem 11.4. However, we can apply Theorem 11.3, which tells us that the corrected solution $(\widehat{s}^T, \widehat{y}^T)^T$ obtained after one step of iterative refinement satisfies

$$\begin{aligned} \left| \begin{bmatrix} b \\ 0 \end{bmatrix} - \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} \widehat{s} \\ \widehat{y} \end{bmatrix} \right| &\le mn\gamma_{cm} \begin{bmatrix} H_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} |\widehat{r}_1| \\ |\widehat{r}_2| \end{bmatrix} + u \begin{bmatrix} I & |A| \\ |A^T| & 0 \end{bmatrix} \begin{bmatrix} |\widehat{s}| \\ |\widehat{y}| \end{bmatrix} \\ &\quad + \gamma_{m+n+1} \left(\begin{bmatrix} I & |A| \\ |A^T| & 0 \end{bmatrix} \begin{bmatrix} |\widehat{s}| \\ |\widehat{y}| \end{bmatrix} + \begin{bmatrix} |b| \\ 0 \end{bmatrix} \right) \\ &\quad + O(u^2). \end{aligned} \tag{19.16}$$

Here, $(\widehat{r}_1^T, \widehat{r}_2^T)^T$ denotes the residual of the augmented system corresponding to the original computed solution. We will make two simplifications to the bound (19.16). First, since $(\widehat{r}_1^T, \widehat{r}_2^T)^T = O(u)$, the first term in the bound may be included in the $O(u^2)$ term. Second, (19.16) yields $|b - \widehat{s} - A\widehat{y}| = O(u)$ and so $|\widehat{s}| \le |A||\widehat{y}| + |b| + O(u)$. With these two simplifications, (19.16) may be written

$$\left| \begin{bmatrix} b \\ 0 \end{bmatrix} - \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} \widehat{s} \\ \widehat{y} \end{bmatrix} \right| \le 2\gamma_{m+n+2} \left(\begin{bmatrix} 0 & |A| \\ |A^T| & 0 \end{bmatrix} \begin{bmatrix} |\widehat{s}| \\ |\widehat{y}| \end{bmatrix} + \begin{bmatrix} |b| \\ 0 \end{bmatrix} \right) + O(u^2).$$

In view of the Oettli–Prager result (Theorem 7.3) this inequality tells us that, asymptotically, the solution $\widehat{y}$ produced after one step of fixed precision iterative refinement has a small componentwise relative backward error with respect to the augmented system. However, this backward error allows the two occurrences of A in the augmented system coefficient matrix to be perturbed differently, and thus is not a true componentwise backward error for the LS problem. Nevertheless, the result tells us that iterative refinement can be expected to produce some improvement in stability. Note that the bound in Theorem 19.2 continues to hold if we perturb the two occurrences of A in the augmented system differently. Therefore the bound is applicable to iterative refinement (with $E = |A|$, $f = |b|$), and so we can expect iterative refinement to mitigate the effects of poor row and column scaling of A. Numerical experiments show that these predictions are borne out in practice [549, 1991].

Turning to mixed precision iterative refinement, we would like to apply the analysis of §11.1, with "$Ax = b$" again identified with the augmented system. However, the analysis of §11.1 requires a backward error result in which only the coefficient matrix is perturbed (see (11.1)). This causes no difficulties because from Theorem 19.4 we can deduce a normwise result (cf. Problem 7.7):

$$\left(\begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} + \Delta\right)\begin{bmatrix} \widehat{r} \\ \widehat{x} \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix}, \qquad \|\Delta\|_2 \le c_{m,n} u \left\| \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \right\|_2.$$

The theory of §11.1 then tells us that mixed precision iterative refinement will converge as long as the condition number of the augmented system matrix is not too large and that the rate of convergence depends on this condition number. How does the condition number of the augmented system relate to that of A? Consider the matrix that results from the scaling $A \leftarrow \alpha^{-1}A$ ($\alpha > 0$):

$$C(\alpha) = \begin{bmatrix} \alpha I & A \\ A^T & 0 \end{bmatrix}. \tag{19.17}$$

Björck [106, 1967] shows that the eigenvalues of $C(\alpha)$ are

$$\lambda(C(\alpha)) = \begin{cases} \dfrac{\alpha}{2} \pm \left(\dfrac{\alpha^2}{4} + \sigma_i^2\right)^{1/2}, & i = 1\colon n, \\ 0, & m - n \text{ times}, \end{cases} \tag{19.18}$$

where σ_i, $i = 1\colon n$, are the singular values of A, and that

$$\sqrt{2}\kappa_2(A) \le \min_\alpha \kappa_2(C(\alpha)) \le 2\kappa_2(A), \quad \max_\alpha \kappa_2(C(\alpha)) > \kappa_2(A)^2, \tag{19.19}$$

with $\min_\alpha \kappa_2(C(\alpha))$ being achieved for $\alpha = \sigma_n/\sqrt{2}$ (see Problem 19.7). Hence $C(\alpha)$ may be much more ill conditioned than A. However, in our analysis we are at liberty to take $\min_\alpha \ \kappa_2(C(\alpha))$ as the condition number, because scaling the LS problem according to $b - Ax \leftarrow (b - Ax)/\alpha$ does not change the computed solution or the rounding errors in any way (at least not if α is a power of the machine base). Therefore it is, $\kappa_2(A)$ that governs the behaviour of mixed precision iterative refinement, irrespective of the size of the LS residual. As Björck [111, 1990] explains, this means that "in a sense iterative refinement is *even more satisfactory* for large residual least-squares problems." He goes on to explain that "When residuals to the augmented system are accumulated in precision β^{-t_2}, $t_2 \geq 2t_1$, this scheme gives solutions to full single-precision accuracy even though the initial solution may have no *correct significant figures.*"

Iterative refinement can be applied with the MGS method. Björck [106, 1967] gives the details and shows that mixed precision refinement works just as well as it does for Householder's method.

19.6. The Seminormal Equations

When we use a QR factorization to solve an LS problem $\min_x \|b - Ax\|_2$, the solution x is determined from the equation $Rx = Q^Tb$ (or via a similar expression involving Q for the MGS method). But if we need to solve for several right-hand sides that are not all available when the QR factorization is computed, we need to store Q before applying it to the right-hand sides. If A is large and sparse it is undesirable to store Q, as it can be much more dense than A. We can, however, rewrite the normal equations as

$$R^TRx = A^Tb,$$

which are called the *seminormal equations*. The solution x can be determined from these equations without the use of Q. Since the cross product matrix A^TA is not formed explicitly and R is determined stably via a QR factorization, we might expect this approach to be more stable than the normal equations method.

Björck [110, 1987] has done a detailed error analysis of the seminormal equations (SNE) method, under the assumption that R is computed by a backward stable method. His forward error bound for the SNE method is of the same form as that for the normal equations method, involving a factor $\kappa_2(A)^2$. Thus the SNE method is not backward stable. Björck considers applying one step of fixed precision iterative refinement to the SNE method, and he calls the resulting process the corrected seminormal equations (CSNE) method:

$$R^TRx = A^Tb$$
$$r = b - Ax$$

$$R^TRw = A^Tr$$
$$y = x + w$$

It is important that the normal equations residual be computed as shown, as $A^T(b - Ax)$, and not as $A^Tb - A^TAx$. Björck derives a forward error bound for the CSNE method that is roughly of the form

$$\frac{\|x-y\|_2}{\|x\|_2} \le c_{m,n}\left(\kappa_2(A)u \cdot \kappa_2(A)^2u\left(1+\frac{\|b\|_2}{\|A\|_2\|x\|_2}\right)+\frac{\kappa_2(A)^2u\|r\|_2}{\|A\|_2\|x\|_2}\right).$$

Hence, if $\kappa_2(A)^2u \le 1$, the CSNE method has a forward error bound similar to that for a backward stable method, and the bound is actually smaller than that for the QR method if $\kappa_2(A)^2u \ll 1$ and r is small. However, the CSNE method is not backward stable for all A.

19.7. Backward Error

Although it has been known since the 1960s that a particular method for solving the LS problem, namely the Householder QR factorization method, yields a small normwise backward error (see $19.2), it was for a long time an open problem to obtain a formula for the backward error of an arbitrary approximate solution. Little progress had been made towards solving this problem until Waldén, Karlson, and Sun [1060, 1995] found an extremely elegant solution. We will denote by $\lambda_{\min}$ and $\sigma_{\min}$ the smallest eigenvalue of a symmetric matrix and the smallest singular value of a general matrix, respectively.

Theorem 19.5 (Waldén, Karlson, and Sun). *Let $A \in \mathbb{R}^{m\times n}$ $(m \ge n)$, $b \in \mathbb{R}^m$, $0 \ne y \in \mathbb{R}^n$, and $r = b - Ay$. The normwise backward error*

$$\eta_F(y) := \min\{\, \|[\Delta A,\ \theta\Delta b]\|_F : \|(A+\Delta A)y-(b+\Delta b)\|_2 = \min\,\} \tag{19.20}$$

is given by

$$\eta_F(y) = \begin{cases} \dfrac{\|r\|_2}{\|y\|_2}\sqrt{\mu}, & \lambda_* \ge 0, \\ \left(\dfrac{\|r\|_2^2}{\|y\|_2^2}\mu + \lambda_*\right)^{1/2}, & \lambda_* < 0, \end{cases}$$

where

$$\lambda_* = \lambda_{\min}\left(AA^T - \mu\frac{rr^T}{\|y\|_2^2}\right), \quad \mu = \frac{\theta^2\|y\|_2^2}{1+\theta^2\|y\|_2^2}.$$

The backward error (19.20) is not a direct generalization of the usual normwise backward error for square linear systems, because it minimizes

Table 19.1. *LS backward errors and residual for Vandermonde system.*

	$\frac{\eta_F}{\|[A,b]\|_F}$ $(\theta = 1)$	$\frac{\eta_F}{\|A\|_F}$ $(\theta = \infty)$	$\|r\|_2$
NE	8.6×10^{-12}	2.40×10^{-10}	6.3×10^{-8}
QR	8.6×10^{-18}	2.41×10^{-16}	6.2×10^{-14}
MGS	2.6×10^{-18}	7.32×10^{-17}	2.0×10^{-14}
MGS(bad)	2.0×10^{-11}	5.66×10^{-10}	5.8×10^{-7}

$\|[\Delta A, \theta \Delta b]\|_F$ instead of $\max\{\|\Delta A\|_2/\|E\|_2, \|\Delta b\|_2/\|f\|_2\}$. However, the Parameter θ allows us some flexibility: taking the limit $\theta \to \infty$ forces $\Delta b = 0$, giving the case where only A is perturbed.

Theorem 19.5 can be interpreted as saying that if $\lambda_* \geq 0$ then the backward error is essentially that given by Theorem 7.1 for a consistent system. If $\lambda_* < 0$, however, the nearest perturbed system of which y is the LS solution is inconsistent. A sufficient condition for $\lambda_* < 0$ is $b \notin \mathrm{range}(A)$ (assuming $\mu \neq 0$), that is, the original system is inconsistent.

The formulae given in Theorem 19.5 are unsuitable for computation because they can suffer from catastrophic cancellation when $\lambda_* < 0$. Instead, the following alternative formula derived in [1060, 1995] should be used (see Problem 19.9):

$$\eta_F(y) = \min\{\eta_1, \ \sigma_{\min}([A \ \ \eta_1 C])\}, \tag{19.21a}$$

$$\eta_1 = \frac{\|r\|_2}{\|y\|_2}\sqrt{\mu}, \qquad C = I - \frac{rr^T}{r^T r}. \tag{19.21b}$$

To illustrate Theorem 19.5, we consider an LS problem with a 25 x 15 Vandermonde matrix $A = (p_j^{i-1})$, where the p_i are equally spaced on [0, 1], and $b = Ax$ with $x_i = i$ (giving a zero residual problem). The condition number $\kappa_2(A) = 1.47 \times 10^9$. We solved the LS problem in MATLAB in four different ways: by using the NE with Cholesky factorization, via Householder QR factorization, and via the MGS method, using both the stable approach described in §19.3 and the unstable approach in which Q^Tb is formed as a matrix–vector product (denoted MGS(bad)). The results, including the norms of the residuals $r = b - A\widehat{x}$, are shown in Table 19.1. As would be expected from the analysis in this chapter, the QR and stable MGS methods produce backward stable solutions, but the NE method and the unstable MGS approach do not.

As we saw in the analysis of iterative refinement, sometimes we need to consider the augmented system with different perturbations to A and A^T. The next result shows that from the point of view of normwise perturbations and componentwise perturbations bounded by $|\Delta A| \le \epsilon G|A|$, the lack of "symmetry" in the perturbations has little effect on the backward error of y.

Lemma 19.6 (Kiełbasiński and Schwetlick). *Let $A \in \mathbb{R}^{m\times n}$ $(m \ge n)$ and consider the perturbed augmented system*

$$\begin{bmatrix} I & A+\Delta A_1 \\ (A+\Delta A_2)^T & 0 \end{bmatrix} \begin{bmatrix} s \\ y \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}.$$

There is a vector $\hat{s}$ and a perturbation ΔA with

$$\Delta A = G_1\Delta A_1 + G_2\Delta A_2, \qquad G_1^T = G_1 = G_1^2, \quad G_1 + G_2 = I, \tag{19.22}$$

such that

$$\begin{bmatrix} I & A+\Delta A \\ (A+\Delta A)^T & 0 \end{bmatrix} \begin{bmatrix} \hat{s} \\ y \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix},$$

that is, y solves the LS problem $\min_x \|(A+\Delta A)x - b\|_2$.

Proof. If $s = b - (A + \Delta A_1)y = 0$ we take $\Delta A = \Delta A_1$. Otherwise, we set

$$\Delta A := P\Delta A_2 + (I - P)\Delta A_1 =: \Delta A_1 + PH,$$

where $P = ss^T/s^Ts$ and $H = \Delta A_2 - \Delta A_1$. We have

$$\hat{s} := b - (A+\Delta A)y = b - (A+\Delta A_1)y - PHy = \beta s,$$

where $\beta = 1 - s^THy/s^Ts$. Then

$$\begin{aligned} (A+\Delta A)^T\hat{s} &= (A+\Delta A_2 + (P-I)H)^T\beta s \\ &= \beta\big((A+\Delta A_2)^Ts + H^T(P-I)s\big) = 0. \qquad \square \end{aligned}$$

Note that (19.22) implies a bound stronger than just $\|\Delta A\|_2 \le \|\Delta A_1\|_2 + \|\Delta A_2\|_2$:

$$\|\Delta A\|_p \le (\|\Delta A_1\|_p^2 + \|\Delta A_2\|_p^2)^{1/2}, \qquad p = 2, F.$$

Turning to componentwise backward error, the simplest approach is to apply the componentwise backward error $\omega_{E,f}(y)$ to the augmented system (19.3), setting

$$E = \begin{bmatrix} 0 & E_A \\ E_A^T & 0 \end{bmatrix}$$

so as not to perturb the diagonal blocks I and O of the augmented system coefficient matrix. However, this approach allows A and A^T to undergo different perturbations ΔA_1 and ΔA_2 with $\Delta A_1 \ne \Delta A_2^T$ and thus does not give a true backward error, and Lemma 19.6 is of no help. This problem can be overcome by using a structured componentwise backward error to force symmetry of the perturbations; see Higham and Higham [527, 1992] for details. One problem remains: as far as the backward error of y is concerned, the vector r in the augmented system is a vector of free parameters, so to obtain the true componentwise backward error we have to minimize the structure-preserving componentwise backward error over all r. This is a nonlinear optimization problem to which no closed-form solution is known. Experiments show that when y is a computed LS solution, $r = b - Ay$ is often a good approximation to the minimizing r [527, 1992], [549, 1991].

19.8. Proof of Wedin's Theorem

In this section we give a proof of Theorem 19.1. We define $P_A := AA^+$, the orthogonal projection onto range(A).

Lemma 19.7. *Let $A, B \in \mathbb{R}^{m\times n}$. If rank(A) = rank(B) and* $\eta = \|A^+\|_2\|A-B\|_2 < 1$, *then*

$$\|B^+\|_2 \le \frac{1}{1-\eta}\|A^+\|_2.$$

Proof. Let r = rank(A). A standard result on the perturbation of singular values gives

$$\sigma_r(B) \ge \sigma_r(A) - \|A-B\|_2,$$

that is,

$$\|B^+\|_2^{-1} \ge \|A^+\|_2^{-1} - \|A-B\|_2 > 0,$$

which gives the result on rearranging. □

Lemma 19.8. *Let $A, B \in \mathbb{R}^{m\times n}$. If* rank($A$) = rank($B$) *then*

$$\|P_A(I-P_B)\|_2 = \|P_B(I-P_A)\|_2 \le \|A-B\|_2 \min\{\, \|A^+\|_2, \|B^+\|_2 \,\}.$$

Proof. We have

$$\begin{aligned}
\|P_B(I-P_A)\|_2 &= \|(P_B(I-P_A))^T\|_2 \\
&= \|(I-P_A)P_B\|_2 \\
&= \|(I-AA^+)BB^+\|_2 \\
&= \|(I-AA^+)(A+(B-A))B^+\|_2 \\
&= \|(I-AA^+)(B-A)B^+\|_2 \\
&\le \|A-B\|_2\|B^+\|_2.
\end{aligned}$$

The result then follows from the (nontrivial) equality $\|P_A(I-P_B)\|_2 = \|P_B(I-P_A)\|_2$; see, for example, Stewart [943, 1977, Thm. 2.3] or Stewart and Sun

[954, 1990, Lem. 3.3.5], where proofs that use the CS decomposition are given. □

Proof of Theorem 19.1. Let $B := A + AA$. We have, with $r = b - Ax$,

$$\begin{aligned} y - z &= B^+(b + \Delta b) - z = B^+(r + Ax + \Delta b) - x \\ &= B^+(r + Bx - \Delta Ax + \Delta b) - x \\ &= B^+(r - \Delta Ax + \Delta b) - (I - B^+B)x \\ &= B^+(r - \Delta Ax + \Delta b), \end{aligned} \tag{19.23}$$

since B has full rank. Now

$$B^+r = B^+(BB^+)r = B^+P_Br = B^+P_B(I - P_A)r. \tag{19.24}$$

Applying Lemmas 19.7 and 19.8, we obtain

$$\begin{aligned} \|B^+r\|_2 &\le \|B^+\|_2(\|B - A\|_2\|A^+\|_2)\|r\|_2 \\ &\le \frac{\|A^+\|_2}{1 - \|A^+\|_2\|\Delta A\|_2}\|\Delta A\|_2\|A^+\|_2\|r\|_2 \\ &= \frac{\kappa_2(A)^2\epsilon}{1 - \kappa_2(A)\epsilon}\frac{\|r\|_2}{\|A\|_2}. \end{aligned} \tag{19.25}$$

Similarly,

$$\begin{aligned} \|B^+(-\Delta Ax + \Delta b)\|_2 &\le \frac{\|A^+\|_2}{1 - \kappa_2(A)\epsilon}\epsilon(\|A\|_2\|x\|_2 + \|b\|_2) \\ &= \frac{\kappa_2(A)\epsilon}{1 - \kappa_2(A)\epsilon}\left(1 + \frac{\|b\|_2}{\|A\|_2\|x\|_2}\right)\|x\|_2 \\ &\le \frac{\kappa_2(A)\epsilon}{1 - \kappa_2(A)\epsilon}\left(2 + \frac{\|r\|_2}{\|A\|_2\|x\|_2}\right)\|x\|_2. \end{aligned} \tag{19.26}$$

The bound for $\|x - y\|_2/\|x\|_2$ is obtained by using inequalities (19.25) and (19.26) in (19.23).

Turning to the residual, using (19.23) we find that

$$\begin{aligned} s - r &= \Delta b + B(x - y) - \Delta Ax \\ &= \Delta b - BB^+(r - \Delta Ax + \Delta b) - \Delta Ax \\ &= (I - BB^+)(\Delta b - \Delta Ax) - BB^+r. \end{aligned}$$

Since $\|I - BB^+\|_2 = \min\{1, m - n\}$,

$$\|r - s\|_2 \le \|\Delta b\|_2 + \|\Delta Ax\|_2 + \|BB^+r\|_2 \le \epsilon(\|b\|_2 + \|A\|_2\|x\|_2) + \|BB^+r\|_2.$$

Using (19.24), Lemma 19.8, and $\|BB^+\|_2 = 1$, we obtain

$$\|BB^+r\|_2 = \|BB^+P_B(I - P_A)r\|_2 \le \|\Delta A\|_2\|A^+\|_2\|r\|_2 \le \kappa_2(A)\epsilon\|r\|_2.$$

Hence

$$\frac{\|r-s\|_2}{\|b\|_2} \le \epsilon \left(1 + \frac{\|A\|_2\|x\|_2}{\|b\|_2} + \kappa_2(A)\frac{\|r\|_2}{\|b\|_2}\right) \le \epsilon(1+2\kappa_2(A)).$$

For the attainability, see Wedin [1069, 1973, §6]. □

Note that, as the proof has shown, Wedin's theorem actually holds without any restriction on m and n, provided we define $x = A^+b$ and $y = (A+\Delta A)^+(b+\Delta b)$ when $m < n$ (in which case $r = 0$). We consider underdetermined systems in detail in the next chapter. The original version of Wedin's theorem also requires only $\mathrm{rank}(A) = \mathrm{rank}(A+\Delta A)$ and not that A have full rank.

19.9. Notes and References

The most comprehensive and up to date treatment of the LS problem is the book by Björck [116, 1996], which is an updated and expanded version of [112, 1990]. It treats many aspects not considered here, including rank-deficient, weighted, and constrained problems. An early book devoted to numerical aspects of the LS problem was written by Lawson and Hanson [695, 1974], who, together with Stewart [941, 1973], were the first to present error analysis for the LS problem in textbook form.

The history of the LS problem is described in the statistically oriented book by Farebrother [363, 1988].

The pseudo-inverse A^+ underlies the theory of the LS problem, since the LS solution can be expressed as $x = A^+b$. An excellent reference for perturbation theory of the pseudo-inverse is Stewart and Sun [954, 1990, §3.3]. The literature on pseudo-inverses is vast, as evidenced by the annotated bibliography of Nashed and Rall [786, 1976], which contains 1,776 references published up to 1976.

Normwise perturbation theory for the LS problem was developed by various authors in the 1960s and 1970s. The earliest analysis was by Golub and Wilkinson [466, 1966], who gave a first-order bound and were the first to recognize the potential $\kappa_2(A)^2$ sensitivity. A nonasymptotic perturbation bound was given by Björck [107, 1967], who worked from the augmented system.

An early set of numerical experiments on the Householder, Gram-Schmidt, and normal equations methods for solving the LS problem was presented by Jordan [618, 1968]; this paper illustrates the incomplete understanding of perturbation theory and error analysis for the LS problem at that time.

van der Sluis [1041, 1975] presents a geometric approach to LS perturbation theory and gives lower bounds for the effect of worst-case perturbations. Golub and Van Loan [470, 1989, Thm. 5.3.1] give a first-order analogue of Theorem 19.1 expressed in terms of the angle θ between b and $\mathrm{range}(A)$ instead of the residual r.